Image-Based Detection and Classification of Property Changes for Pre-Inspection Assessment

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1 Introduction

In housing corporations like De Goede Woning Apeldoorn, inspections before and after tenant occupancy are still manual and require on-site visits. The rental department co-worker must enter the property to detect changes such as damage, alterations, or maintenance needs. Like a hole in the door, or wear and tear. This process is time-consuming and difficult to scale. Even with the guidelines, the 10 rental department colleagues react differently to damage. Visual documentation, such as photographs, is archived for every house in our company but can not be analyzed inside the property. There, for the rental department needs/wants an assessment method that helps them going into the house well prepared and even help them avoid a discussion with the tenant. There will be less waist of time and tenants get an report by forehand on what changes they need to make before moving out.

Recent developments in computer vision have enabled the detection of visual differences, object classification, and scene understanding in various domains, mainly medical imaging, infrastructure monitoring, garbage recognition, and manufacturing. However, image-based inspection automation in the rental housing sector has not yet been explored. Currently, there is no automated method that compares pre- and post-lease images to identify property changes before a physical inspection is scheduled. This a scientific and practical opportunity. If a model could detect and classify visual changes from photographs, it could assist inspections by prioritizing which properties need attention and reducing time and even unnecessary visits and discussions. At the same time, the domain introduces specific challenges such as inconsistent lighting, camera angle variability, and differing interior layouts.

Problem Statement

Housing corporations like De Goede Woning want the opportunity to make property inspections more efficient and scalable. Instead of relying only on scheduled visits or tenant reports before they move out, an image based approach like computer vision can provide improved insight into the condition of a property over time. This creates new possibilities like a self-service tool where tenants can easily perform a check before moving out, or where rental colleagues can assess a property from their desk by the pictures of the client. By standardizing this process, inspections become not only more reliable and informed, but also more user friendly for both tenants and the housing corporations.

Key Research Question

To what extent can AI tooling, such as computer vision, be used within business operations to compare images and accurately detect differences?

Sub-Research Questions

- Which computer vision techniques are most suitable for comparing time-separated property images?
- How can visual changes be categorized into classes such as damage, alterations, wear and tear or maintenance needs?
- How can computer vision models detect property changes using lease images?
- How can issues like lighting, camera angle, paint, wallpaper or image quality be mitigated?

2 Seminal Papers

This project builds on existing research in automated visual inspection and change detection. Several works inform the scientific foundation of this proposal.

[1] Cheng provides a comprehensive review of deep learning-based change detection across visual domains, highlighting the importance of temporal comparison and feature extraction for detecting structural variations in images. Their work establishes the relevance of computer vision models for monitoring change over time.

[2] Kang demonstrate how image-based automatic defect detection can be integrated with building information models to assess deteriorated structures. Their approach shows how visual inspection tasks can be automated in the built environment, reinforcing the applicability of image-based classification to physical infrastructure.

In a related urban context, Maarten Sukel[3] applied object detection techniques to identify garbage in public environments. This work demonstrates that vision-based monitoring can operate in real-world, high-variability settings, supporting the feasibility of visual classification and anomaly detection beyond controlled laboratory data.

These works collectively inform the methodological direction of this project by demonstrating that visual change detection, automatic classification, and environmental monitoring can be achieved with modern computer vision models. However, none of these studies explore automated comparison of pre- and post-lease property images for housing inspections, leaving a clear research gap that this thesis addresses.

3 Objectives & Goals

Project Goal

The goal of this project is to design and evaluate a computer vision approach that can detect and classify visual changes in rental

property images to support inspection decision making and assist housing staff during the leasing process.

Objectives

- Identify relevant computer vision methods for visual change detection.
- Collect or simulate a dataset of pre- and post-lease property images.
- Develop or adapt a model to compare and classify visual changes.
- Define meaningful change categories (e.g., damage, maintenance needs, alterations).
- Introduce a scoring or severity assessment that indicates urgency or inspection priority.
- Evaluate the performance of the model using appropriate metrics.
- Provide outputs that help housing staff decide whether an inspection is necessary before entering a property.
- Reflect on limitations and potential application in real-world housing inspections and leasing workflows.
- Introducing a benchmark for identifying damage, maintenance, wear and tear.

Extended Goal Breakdown

(1) **Develop an Image-Based Change Detection Model**
Objective: Create a computer vision model that scores property images taken before and after leasing periods.
(2) **Classify Types of Visual Changes**
Objective: Distinguish between categories such as damage, alterations, wear and tear, and maintenance needs.
(3) **Introduce a Change Scoring or Severity Mechanism**
Objective: Provide a measurable score or classification level to prioritize follow-up actions.
(4) **Support Inspection Evaluation**
Objective: Enable property and housing staff to determine whether a physical inspection is needed before visiting the property.
(5) **Develop a Proof of Concept**
Objective: Build and test a prototype that demonstrates feasibility using sample or real property images.

4 Project Plan

The project is divided into four phases aligned with the thesis milestones:

Phase 1: Thesis Design (January–February)

- Define research question, scope, and objectives.
- Conduct an initial literature review on change detection and computer vision.
- Identify relevant data sets or image sources.

Phase 2: Review of the literature / exploratory work (March)

- Extend the literature review based on state-of-the-art methods.
- Refine research sub-questions.
- Explore dataset preparation and pre processing requirements.

Phase 3: Methodology Draft (Mid-April)

- Select one or more baseline computer vision techniques.
- Define feature extraction, comparison strategy, and evaluation metrics.
- Draft experimental setup and validation design.

Phase 4: Early Prototype and Draft Thesis (May–June)

- Develop a working prototype.
- Evaluate results on a subset of the data.
- Write draft thesis sections based on progress.

A short table-style view:

Phase	Timeframe	Output
1	Jan–Feb	Thesis Design (4 pages)
2	March	Literature Review (3 pages)
3	Mid-April	Methodology Draft (3 pages)
4	May–June	Prototype + Draft Thesis

Roadmap

Sprint	**Dates**	**Focus / Deliverable**
Sprint 1	Jan 5 – Jan 18	Start Thesis Design (RQ, scope, intro)
Sprint 2	Jan 19 – Feb 1	Continue Thesis Design
Sprint 3	Feb 2 – Feb 15	Finish Thesis Design (buffer before deadline)
Sprint 4	Feb 16 – Mar 1	Literature Review + Identify relevant datasets
Sprint 5	Mar 2 – Mar 15	Continue Lit Review + Begin data collection
Sprint 6	Mar 16 – Mar 29	Finish Lit Review (buffer) + Finalize dataset pla
Sprint 7	Mar 30 – Apr 12	Draft Methodology + Data preparation
Sprint 8	Apr 13 – Apr 26	Define evaluation metrics + model selection (bu
Sprint 9	Apr 27 – May 10	Creating the model (PoC)
Sprint 10	May 11 – May 24	Test PoC + refine model (buffer week)
Sprint 11	May 25 – Jun 7	Full Draft Thesis + integrate model results
Sprint 12	Jun 8 – Jun 21	Final working version of thesis + polish finding
Sprint 13	Jun 22 – Jun 30	*Buffer only — no new work*

References

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[3] Maarten Sukel. 2019. *Automated Garbage Detection in Urban Environments using Deep Learning*. Master's thesis. University of Amsterdam. https://github.com/maartensukel